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**WMO Data Interoperability,
Standardization and Exchange**

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**WORLD
METEOROLOGICAL
ORGANIZATION**

What is Interoperability?



Why interoperability?

Weather Station → Proprietary Format → **×** Hydrological System

Interoperability

Aspect of interoperability	Hydrometeorological example
Technical interoperability	The meteorological agency's system sends rainfall and temperature data to the water agency through an API or web service.
Syntactic interoperability	Both systems use the same data format, such as JSON, XML, or CSV, so the receiving system can read the data.
Semantic interoperability	Both agencies agree that rainfall = accumulated precipitation in millimeters (mm) and temperature is measured in degrees Celsius (°C).
Organizational interoperability	The agencies agree on which data to share, how often to share it, and who is responsible for maintaining the data.
Legal/regulatory interoperability	Data sharing follows relevant government policies, data-sharing agreements, and regulations.

What is Standards



Why Standards?

Agency A → proprietary format → **✗** Agency B cannot easily use the data

Open Standards

Share, Integrate,
Understand, And Reuse Data
Efficiently

Reason	How it helps in Hydrometeorology	Example
Interoperability	Allows different systems to exchange data	Weather station → Hydrological database
Data sharing	Makes it easier for agencies to share weather and water data	Meteorological agency shares rainfall data with water authority
Avoid vendor lock-in	Reduces dependence on one software or equipment provider	Data can be used with different software
Common understanding	Ensures data has consistent meaning, units, and definitions	Rainfall = 50 mm/24 hours
Integration	Combines data from stations, satellites, radar, and models	Satellite + rain gauge + river gauge data
Disaster management	Supports timely flood and drought warnings	Rainfall data used for flood forecasting
Data reuse	Makes data easier to access and use in the future	Historical climate data reused for research
Cost efficiency	Reduces the need for custom solutions between systems	One standard API can serve multiple applications

What is Open-Source?



Open source: A program can be open source but have paid versions/services

- **Free** can mean you can use something without paying money. It may still be closed-source.
- **Open source** means the **source code is available** under a license that allows people to inspect, modify, and usually redistribute it.

Source: https://license.md/is-free-software-the-same-as-open-sourcesoftware/#google_vignette

Why open source?

Weather station → Proprietary software → High license cost → Limited customization

Why Open-Source?

Reason	Benefit in Hydrometeorology
Lower cost	Reduces software licensing costs
Flexibility	Can be customized for specific hydrometeorological needs
Interoperability	Works with different systems and open standards
Transparency	Source code can be inspected and verified
Community support	Large communities contribute improvements and solutions
Avoid vendor lock-in	Reduces dependence on one software company
Continuous development	Software can be improved by many developers
Data integration	Makes it easier to combine different data sources
Capacity building	Helps organizations develop local technical skills
Long-term sustainability	Software and data can remain usable without depending on a commercial license

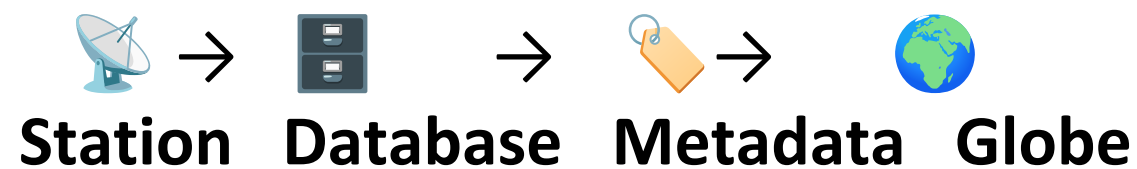
What is Free and Open-Source?



Source: <https://www.mona.uwi.edu/mits/free-and-open-source-software>

Purposes of Metadata

WIGOS WMDR1	WIS2-WCMP2
Describes observations, stations	Describes data products
ISO 19156, includes ISO19115	ISO 19115
XML encoding	GeoJSON encoding
Supports observation metadata	Supports discovery, access, retrieval
WMDR 2 (WCMP2 extension, GeoJSON encoding)	WCMP1: Formalized using ISO19115 metadata standard, encoding: XML



WMDR and WCMP together form the basis for interoperable hydrological data exchange.

WIS2 in a box is Free and Open

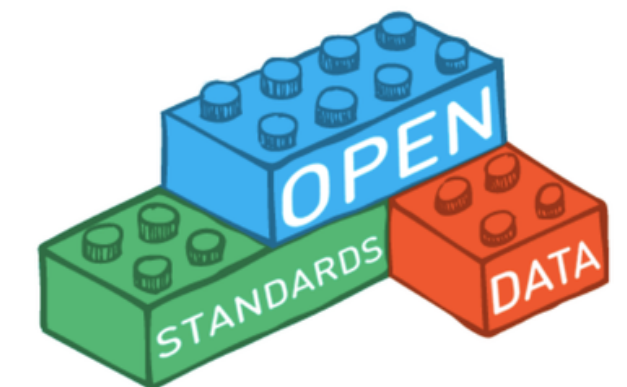
Free and Open Source Software



Open Standards



- MQTT
- GeoJSON
- OGC APIs



Standards Supporting Hydrology

Standards Enabling Interoperability

Applications

WHOS Services

WaterML 2.0 and TSML2.0

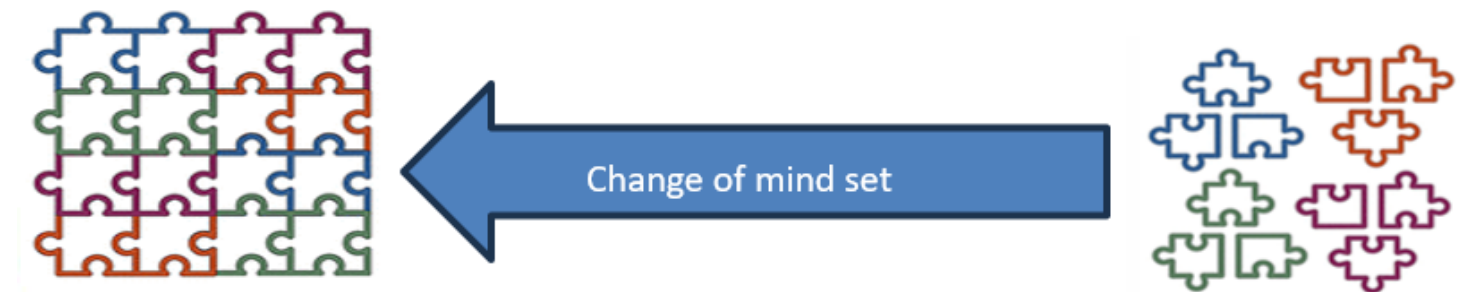
OGC API, STA V2, OMS

ISO Metadata

WIS2

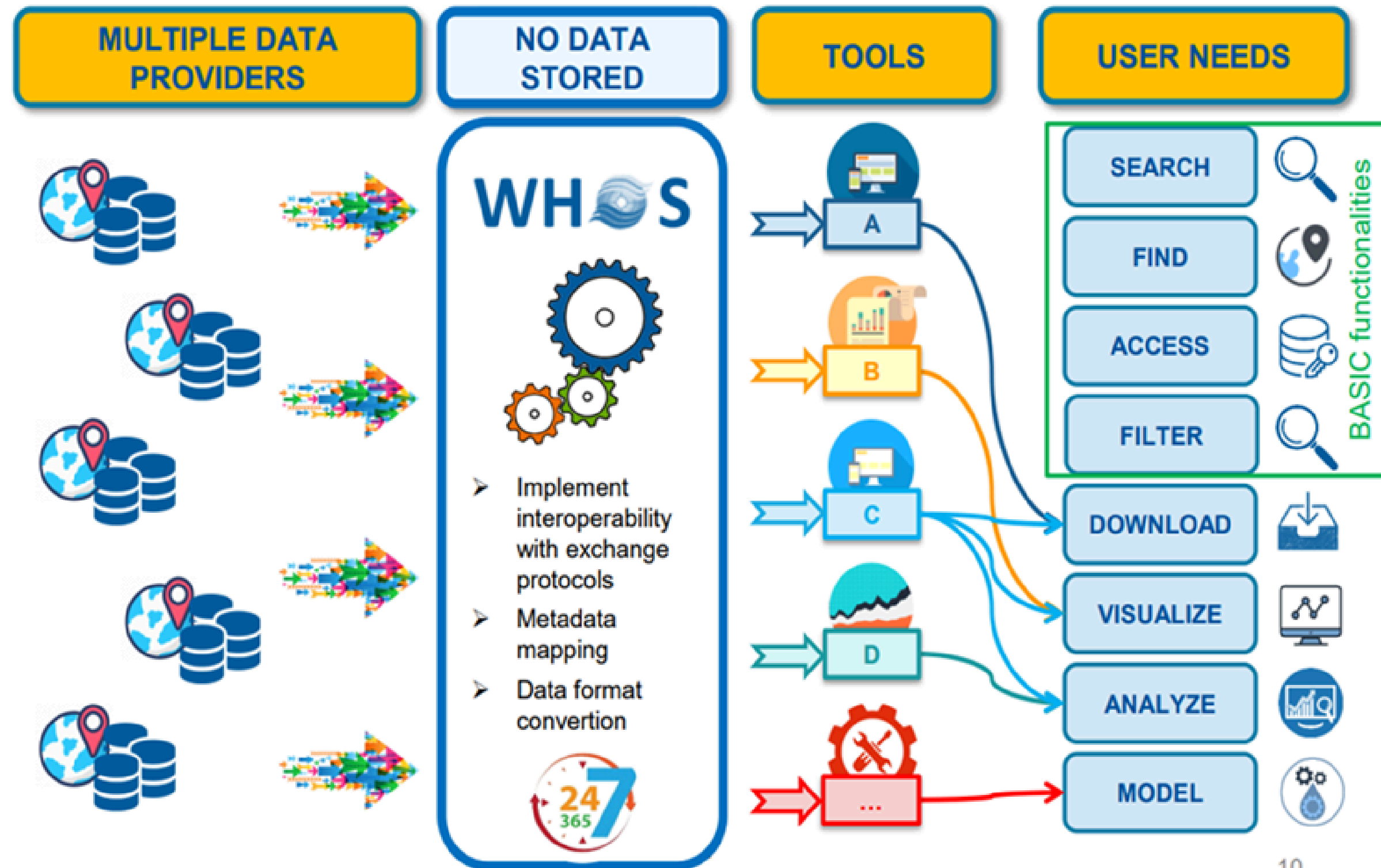
WIGOS

National Hydrological Databases



International standards ensure that hydrological information can be exchanged consistently across organizations and countries

Interoperability in Hydrology



Thank you

Let's work together to make hydrological data discoverable, accessible, and interoperable.

Experts, Contributors and Partners

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